

INTERNSHIP REPORT

*Comparison of Helixer Gene Predictions
Against Expert-Curated Annotations
in Plant Genomes of GBOT*

Author:

Akshay KHOKAN

Supervised by :

Carène RIZZON

Franck SAMSON

Véronique BRUNAUD



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ABBREVIATIONS

CDS: Coding DNA Sequence

FASTA: File format used to store nucleotide or protein sequences

GFF: General Feature Format

GBOT: Genome Browser and Orthologs Tracker

TEs: Transposable Elements

TAIR: The Arabidopsis Information Resource

BLASTP: Protein Basic Local Alignment Search Tool

RNA-seq: RNA sequencing

LaMME: Laboratoire de Mathématiques et Modélisation d'Evry

INRAE: Institut national de recherche pour l'agriculture, l'alimentation et l'environnement

IPS2: Institut des Sciences des Plantes de Paris-Saclay

NAM-5.05: Genome assembly version of *Zea mays* B73 line used in GBOT

Araly v1.0.12: Genome assembly version of *Arabidopsis lyrata* used in GBOT

ARAPORT11: Genome annotation version of *Arabidopsis thaliana* used as a reference

Helixer: Deep learning-based tool for eukaryotic gene prediction

UniProt: Universal Protein Resource

KEYWORDS

Gene prediction, Helixer, structural annotation, transposable elements, comparative genomics.

Comparison of Helixer Gene Predictions Against Expert-Curated Annotations in Plant Genomes of GBOT (Genome Browser & Ortholog Tracker).

Akshay KHOKAN ¹, Carène RIZZON ², Franck SAMSON ³ & Véronique BRUNAUD ³

¹ Master's Student (Year 1) in GENIOMHE-AI, Paris-Saclay University

² Professor, Évry-Paris-Saclay University, LAMME - Supervisor of the internship

³ Research Engineer, INRAE – Co-supervisor of the internship

INTRODUCTION

High-throughput sequencing has revolutionized our ability to read and study genomes. As a result, the number of available plant genome sequences has grown rapidly. Today, thousands of plant genomes are publicly available¹, including those of model plants like *Arabidopsis thaliana* and important crops such as rice, maize, and wheat. One of the main goals of *L'Institut national de recherche pour l'agriculture, l'alimentation et l'environnement* (INRAE), the French national research institute dedicated to agriculture, food, and the environment, is to understand how plants grow, adapt, and respond to their environment. Within INRAE, *L'Institut des Sciences des Plantes de Paris-Saclay* (IPS2) studies the genetic and molecular mechanisms that control plant growth, using a multidisciplinary approach that combines genomics, bioinformatics, and genetics. Therefore, it is essential to correctly identify and annotate all the elements present in these genomes.

Structural annotation is a critical step because errors at this stage can affect all downstream analyses, such as gene expression studies or evolutionary comparisons. Despite many advances, gene prediction remains difficult, especially in plant genomes. These genomes are often large and complex. They contain a high proportion of transposable elements², many repeated sequences, and duplicated genes. In addition, some plant species have undergone polyploidy events, meaning that they possess multiple complete sets of chromosomes, which may originate from the same or different species. All these characteristics make it difficult to correctly identify gene boundaries and structures. As a result, automatic gene prediction methods can produce errors such as missing genes, fragmented genes, or incorrect exon-intron structures. Traditionally, genome annotation relies on expert curation.

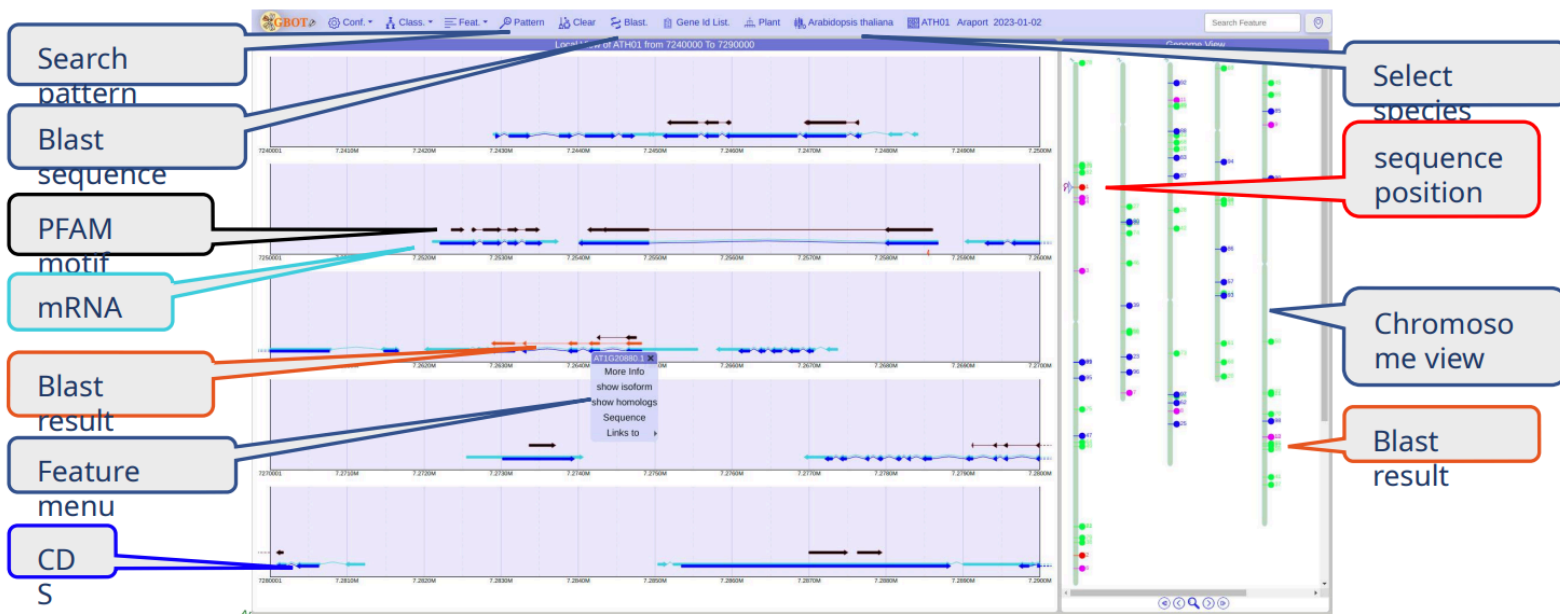


Figure 1 : GBOT Web Interface

GBOT includes multiple tools dedicated to genome visualization, annotation comparison, sequence analysis, and comparative genomics across species :

- Search pattern:** Allows you to search for a specific sequence in the genome.
- Blast sequence:** Launches a BLAST search based on a given sequence.
- PFAM motif:** Displays the protein domains detected in the genes.
- mRNA:** Visualises the annotated mRNAs in the selected region.
- Blast result:** Shows the aligned results of the BLAST search.
- Feature menu:** Provides access to the various available annotations.
- CDS:** Displays the coding sequences of the genes.
- Select species:** Choose the species to display for comparison.
- Sequence position:** Indicates the exact position of the genomic sequences.
- Chromosome view:** Displays chromosome alignments between species.
- BLAST result (comparative):** Shows BLAST alignments between homologous genes from different species.

This approach combines computational predictions with experimental data, such as RNA sequencing, and similarity with known genes from other species. Tools such as Augustus³ are commonly used to perform *ab initio* gene prediction. These predictions are then refined using additional evidence and manual curation by experts. Although this method can produce high-quality annotations, it is slow, expensive, and requires significant expertise. Consequently, many genomes, especially in non-model species, remain poorly annotated and have to be corrected, as is the case for *Arabidopsis lyrata*⁴. Indeed, it is a sister species of the model species *Arabidopsis thaliana* while remaining less extensively studied and annotated, it represents an interesting case for evaluating and improving automatic gene prediction methods through comparative analyses.

To overcome these limitations, new approaches based on machine learning and deep learning have been developed. These methods aim to learn patterns directly from genomic sequences without relying heavily on external data. Tools such as Tiberius⁵ and Helixer⁶ represent this new generation of gene prediction methods. They use neural networks trained on large datasets to identify gene structures directly from DNA sequences. Helixer, in particular, is a deep learning-based tool designed for eukaryotic gene prediction. It uses neural network models to detect coding regions and gene structures in genomic sequences. One of its main advantages is that it can work directly from raw DNA sequences, without requiring extensive additional data, especially useful for newly sequenced genomes or species with limited experimental data. However, despite these advantages, the performance of Helixer is not always consistent across species. Its accuracy can depend on factors such as genome size, complexity, and similarity to the data used during training. Therefore, it is important to evaluate its predictions carefully by comparing them to high-quality reference annotations.

The Genome Browser and Orthologs Tracker (GBOT)⁷ provides a useful framework for this type of analysis. GBOT is a platform that integrates genomic data from different organisms, including bacteria, *Drosophila* species, and plants. It allows users to visualize genome annotations and to perform comparative analyses. It also includes tools to explore orthologous genes between species (*Figure 1*), which is important for genome annotation because conserved genes are more likely to be functional and biologically relevant. The identification of orthologs can therefore help validate predicted genes, transfer functional annotations, and detect potential annotation errors. In a previous study conducted within the GBOT framework, Helixer was applied to four plant genomes among the six in the database.

Feature type	Araport11	TAIR12
Protein-coding genes	27,562	26,867
mRNAs	48,265	36,573
Pseudogenes (transcribed/other)	926/3917	2,140
Repeat regions (# of repeats)		
centromeric_repeat	not annotated	65,718
satellite_DNA	not annotated	11,123
telomeric_repeat	not annotated	4,218

Retrotransposons		
Copia_LTR_retrotransposon	1781	1,449
Gypsy_LTR_retrotransposon	4181	3,550
L1_LINE_retrotransposon	1366	876
LINE_element	81	419
LTR_retrotransposon	not annotated	951
SADHU retrotransposon	8	not annotated
SINE_element	131	389
tRNA_SINE_retrotransposon	not annotated	202
Transposons		
CACTA_TIR_transposon (DNA/En-Spm)	941	704
DNA_transposon	1829	772
hAT_TIR_transposon (DNA/HAT)	1035	841
helitron (RC/helitron)	12945	3,666
Mutator_TIR_transposon (DNA/MuDR)	5410	2,375
PIF_Harbinger_TIR_transposon (DNA/Harbinger)	379	379
Tc1_Mariner_TIR_transposon (DNA/Mariner, DNA/Tc1, DNA/Pogo)	590	648

Table 1 : Comparison of annotated entities in Araport11 and TAIR12. ⁸

The green background highlights values that decreased in the updated version, while the gray background indicates values that increased or were newly introduced.

The predicted annotations were compared with expert-curated annotations in order to evaluate the performance of the tool. This study showed that Helixer has good potential for gene prediction, but also highlighted some limitations, such as errors in gene structure or missing genes in complex regions. These results suggest that further evaluation is needed on additional species.

In this project, we aim to extend this previous work by analyzing new plant genomes. Helixer will be applied to the genomes of *Zea mays*² and *Arabidopsis lyrata*⁴. These two species present different levels of genomic complexity. *Zea mays* has a large and highly repetitive genome, while *Arabidopsis lyrata* is closer to a model organism but still less well annotated than *Arabidopsis thaliana*. The predictions generated by Helixer will be compared to expert-curated annotations in order to evaluate their accuracy. The comparison will focus on several aspects, including the number of predicted genes, the structure of genes, and the agreement between predicted and reference annotations.

Another important aspect of this project is the analysis of annotation versions. A new version of the *Arabidopsis thaliana* annotation (TAIR12)⁹, has recently been released (Public database release: February 2026). The previous one, ARAPORT11¹⁰ (Official release : June 2017), was used in the previous GBOT analysis. Although ARAPORT11 represented a major improvement in the annotation of the *Arabidopsis thaliana* Col-0 genome, it remains based on older genome assemblies and automated annotation pipelines relying mainly on short-read RNA-seq data. Since its release, advances in sequencing and annotation technologies have enabled more accurate and complete genome references. In this context, TAIR12 was developed using improved long-read genome assemblies of the reference ecotype Col-0, including newly resolved nucleolar organizer regions (NORs) located at the tips of chromosomes 2 and 4¹¹, leading to a more complete representation of the genome. In addition to refining protein-coding gene models, TAIR12 improves the representation and annotation of complex genomic regions, including TEs and regions that were not previously annotated in ARAPORT11 : centromeric regions, telomeric regions and satellite DNA (*Table 1*).

The availability of these two versions allows the evaluation of how improvements in reference annotations affect the assessment of the Helixer tool. Comparing Helixer predictions with ARAPORT11 and TAIR12 may also reveal annotation inconsistencies and regions with significantly modified gene models.

In this context, this internship was carried out under the supervision of Carène RIZZON at *Laboratoire de Mathématiques et Modélisation d'Evry* (LaMME), and co-supervised by Véronique BRUNAUD, and Franck SAMSON, both research engineers at INRAE.

SCIENTIFIC QUESTIONS AND OBJECTIVES

The main goal of this project is to evaluate the performance of Helixer for gene prediction in plant genomes. To do this, several scientific questions are asked:

- How accurate are the gene predictions made by Helixer on different plant genomes?

Helixer will be applied to *Zea Mays* and *Arabidopsis Lyrata*, then the predictions will be compared to the expert-curated annotations at exon level, and coding sequence (CDS) level, with the comparison tool built in the previous study. The number of predicted genes, their lengths, and exon structures will be compared. Overlaps between Helixer predictions and reference annotations will be identified in order to distinguish correctly predicted genes, partially predicted genes, and missing genes. The differences will be classified into false positives, false negatives, and structural errors. For the false positives, protein sequences will be generated from their CDS, then a specific analysis will be done on these sequences.

- Does the quality of the reference annotation (more precisely ARAPORT11 vs TAIR12) influence the evaluation of Helixer results?

A specific analysis will be conducted using *Arabidopsis thaliana*. Helixer predictions will be compared separately to the ARAPORT11 and TAIR12 annotations. Specific cases will be analyzed where a gene is present in one annotation version but not in the other, and where Helixer predictions match only one of the two versions.

- How does Helixer annotate bigger genomes with a high proportion of TEs (*Zea Mays*)?

A focused analysis will be conducted on *Zea mays*. The quality of Helixer predictions will be compared between regions rich in TEs and regions with lower repeat content. In TE-rich regions, particular attention will be given to the presence of false positives, where Helixer may incorrectly annotate TEs as genes.

In contrast, in regions with lower repeat content, the analysis will focus on Helixer's ability to correctly identify known genes and avoid false negatives.

To answer these questions, the project has the following objectives:

- Apply Helixer to the genomes of *Zea mays* and *Arabidopsis lyrata*
- Compare and study the differences between Helixer predictions and reference annotations
- Compare Helixer results obtained using ARAPORT11 and TAIR12 annotations for *Arabidopsis thaliana*

METHODOLOGICAL APPROACHES

1. Data used

The data used in this project come from the GBOT database. This database contains genomic data from different organisms. The study mainly focuses on the following species:

- *Zea mays* B73 (NAM-5.05)
- *Arabidopsis lyrata* (Araly v1.0.12)
- *Arabidopsis thaliana* (with ARAPORT11 and TAIR12 annotations)

For each species, the genome sequence and the expert-curated annotations are used. These data are in FASTA and GFF formats. The reference annotations for *Zea Mays*, for *A. Lyrata* and ARAPORT11 are obtained from the Phytozome¹ database.

2. Gene prediction with Helixer

Helixer is used to predict gene structures from genomic sequences. Indeed, the input file is a FASTA file (genome sequence) and the output file is a GFF file (gene predictions). Helixer uses deep learning to identify genes, exons, and introns directly from DNA sequences. This method does not require a lot of additional data, which makes it useful for new or poorly annotated genomes.

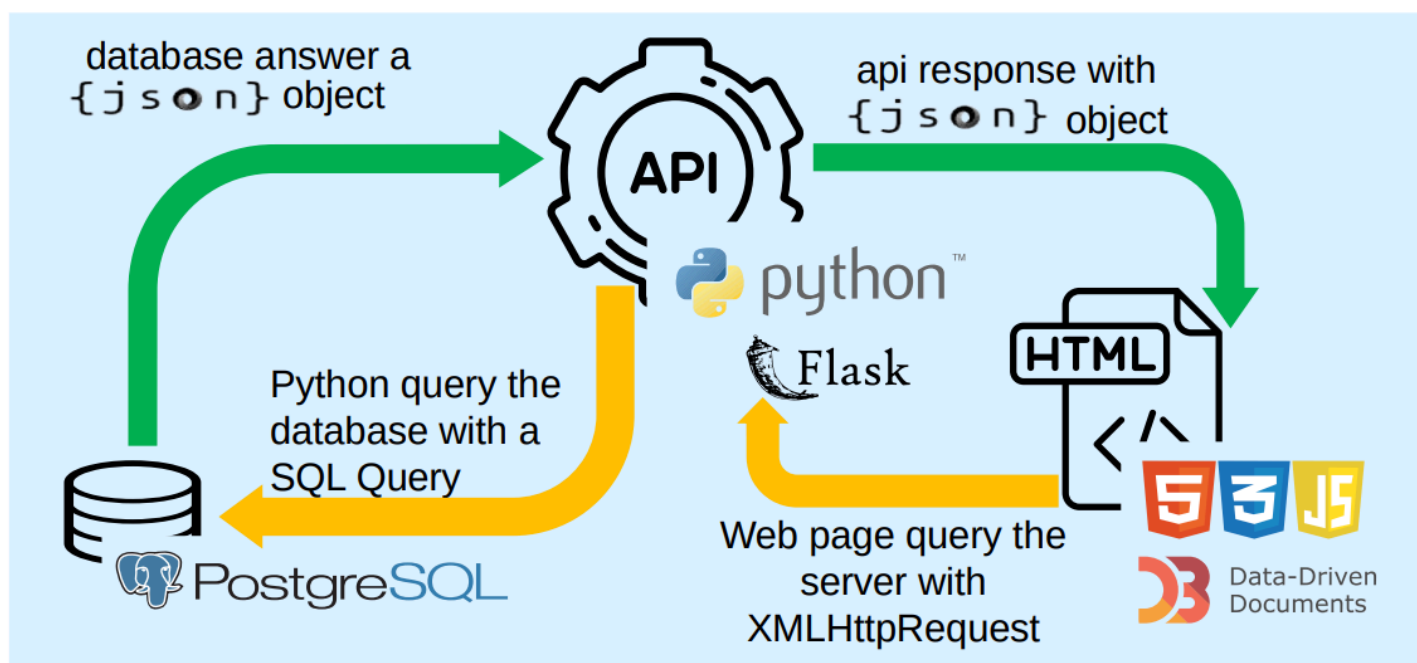


Figure 2 : Functional diagram of GBOT.

The web interface uses JavaScript exclusively for the presentation layer, connected to a PostgreSQL database that enables the storage and querying of genomic data. A layer coded in Python acts as an intermediary between the client's browser and the database server via an API. The web interface, which allows the visualisation of homology relationships within gene families across multiple species, is based on WASM and displays database queries via .json outputs.

However for *Zea Mays*, big computational resources will be needed, given its genome size. Therefore, Helixer will be run on usegalaxy.fr¹², a web platform providing access to bioinformatics tools and computational resources.

3. Comparison of annotations

To evaluate Helixer, its predictions are compared with expert-curated annotations. The comparison is done at exon level and CDS level. For this purpose, a dedicated comparison tool named HelixFlor was developed during the previous study conducted within the GBOT framework. It is still under development and it will be released on [the HelixFlor GitHub repository](#). After the comparison, differences between annotations are analyzed. The HelixFlor tool will detect genes predicted by Helixer but not present in reference annotations (false positives), genes present in reference annotations but not predicted by Helixer (false negatives), and errors in gene structure (wrong exon boundaries, missing exons, etc.). HelixFlor will also highlight the length, the number of exons of the false positives, along with their functional annotation.

A specific analysis is also done for *Arabidopsis thaliana* to compare results obtained with ARAPORT11 and TAIR12. BED files for the false positives, false negatives and true positives will be generated by the comparison tool, for further visualization and manual inspection of the predicted annotations. The protein sequences translated from the CDS of the false positives will be analyzed using Uniprot¹³ and BLASTP¹⁴ to evaluate the coherence and biological relevance of the predicted genes.

5. Data exploration with GBOT

GBOT is designed to explore gene and protein homology relationships. It allows the study of homologous gene families through visualization and the prediction of orthologs. With this database, the use of SQL queries along with Python code will be done to extract and compare data with the comparison tool (*Figure 2*). Moreover, the visualization allows the manual check of the comparison results, and confirms whether the comparison tool performed well on both annotations.

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SUMMARY

This project aims to evaluate the performance of the Helixer tool for gene prediction in plant genomes using data available in the Genome Browser and Orthologs Tracker (GBOT). The first phase focuses on the genomes of *Arabidopsis lyrata* and *Zea mays*, for which Helixer will be used to generate structural gene annotations directly from genomic sequences. The predicted annotations will then be compared to expert-curated annotations in order to assess the accuracy of Helixer predictions at the gene, exon, and coding sequence levels. Particular attention will be given to the identification of false positives, false negatives, and structural prediction errors.

Building on this, the second phase focuses on the comparison between different versions of *Arabidopsis thaliana* reference annotations, namely ARAPORT11 and the recently released TAIR12 annotation (*February 2026*). The objective is to evaluate how updates in reference annotations influence the assessment of Helixer performance and to determine whether Helixer predictions are more consistent with the newer annotation version. Comparative analyses between annotation versions will help identify differences in predicted gene structures and annotation completeness.

Finally, a focused analysis will be conducted on *Zea mays*, a genome characterized by a high proportion of transposable elements and repetitive sequences. In this context, the project will investigate how genome complexity affects Helixer predictions by comparing TE-rich regions with regions containing lower repeat content. The study will evaluate whether repetitive regions increase the occurrence of false positive predictions or lead to missed genes. In addition, genes predicted by Helixer but absent from expert-curated annotations will be analyzed in order to identify potential novel annotations and further evaluate the strengths and limitations of deep learning-based genome annotation approaches.